

Hadrian Fratarcangeli

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Portfolio: ian-fratarcangeli.github.io
GitHub: github.com/Ian-Fratarcangeli

PROFESSIONAL PROFILE

- Strong statistics and machine learning foundation with hands-on experience in quantitative finance and trading.
- Designed deep learning architectures for market prediction, applied change point detection to identify distributional shifts in financial data, and engineered real-time trading engines with risk management controls.
- Interested in the intersection of machine learning and finance, and designing models for market analysis.

EDUCATION

Stony Brook University

MS in Applied Mathematics and Statistics

Stony Brook, NY

August 2025 – December 2026

- Relevant Coursework: Machine Learning in Finance, Financial Mathematics, Stochastic Calculus, Computational Finance, Portfolio Theory, Quantitative Risk Management, Time Series Analysis

Hamilton College

BA in Mathematics and Computer Science

Clinton, NY

August 2021 – May 2025

- Relevant Coursework: Statistical Modeling, Probability, Programming, Statistical Machine Learning, Data Science
- Leadership: Hamilton Football Captain (2023-2024), Psi Upsilon Fraternity President (2023-2024)

WORK EXPERIENCE

Data Annotation

AI Expert Trainer (Part-time Contractor)

Remote

May 2025 – Present

- Generate coding prompts for Large Language Models (LLMs) coupled with complex codebases for testing model correctness, instruction execution, and efficiency.
- Produce coding prompt solutions and corresponding evaluation reports for LLM training.

Oppenheimer & Co. Inc.

Informational Technology Intern

New York, NY

June 2024 – August 2024

- Developed and optimized backend endpoints for the financial advisor platform, enhancing system performance and user experience.
- Implemented unit tests to ensure the reliability and functionality of new and existing code.
- Wrote SQL queries to retrieve and analyze data from company databases, supporting backend development and ensuring data accuracy.
- Led winning intern team on an equity research project, analyzing market data to develop a diversified portfolio that we presented to a board of managing directors.

Hamilton College

Computer Science Teaching Assistant

Clinton, NY

August 2022 – May 2025

- Tutored students in understanding and completing coding labs.
- Provided guidance on programming concepts, debugging, and problem-solving.

RECENT PROJECTS

Trading Engine Simulator

- Performed backtesting including visuals and significant performance metrics with comparisons to market benchmarks.
- Implemented a machine learning driven risk manager to lower drawdowns by assessing regime changes.
- Integrated with Alpaca API for historical asset data to simulate trades with bid-ask spread and slippage.

LSTM-MLP Option Pricer

- Built a neural network with PyTorch to improve option value forecasting by integrating sequential and dense layers.
- Preprocessed option data using Python (pandas) to streamline workflow for static and encoded features.

CUSUM Change Point Detection Algorithm

- Constructed a volatility spike detector using online CUSUM methodology with kernel and likelihood methods.

SKILLS

Technical Skills: Machine Learning, Algorithmic Trading, Statistical Modeling, Risk Management, Data Analysis

Programming Skills: Python, R, C++, SQL, Ruby, Haskell

Software Tools: Git, Excel, Jupyter Notebook, MongoDB, L^AT_EX